

STA 35C: Statistical Data Science III

Lecture 5: Simple Linear Regression

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Announcements

Homework 2: Due Thu, Oct 8, 11:59 PM PT

- Please submit on time and follow the submission instructions
 - To Canvas/Gradescope in a single PDF
 - Email submissions are not accepted
- Feel free to ask questions during lecture, in office hours or post them on Piazza

Agenda

Last time: Statistical learning overview

- *Definition:* A set of tools for understanding data and making informed predictions
- *Goal:* Learn an estimate $\hat{f} : \mathcal{X} \rightarrow \mathcal{Y}$ that
 - (1) Has low prediction error on new data (i.e., has small reducible error $\hat{f}(X) - f(X)$), and
 - (2) Is interpretable enough for the task
- Categorization and tradeoffs:
 - Supervised vs. unsupervised learning
 - Prediction vs. inference
 - Parametric vs. non-parametric methods

Today:

- Regression vs. classification
- (Simple) linear regression
- Least squares, inference, and model assessment

Recap: Supervised vs. unsupervised learning

Most statistical learning problems fall into two categories: supervised or unsupervised

In **supervised learning**:

- Each predictor observation x_i is accompanied by a response y_i
- “Supervised” because the responses guide (supervise) the analysis
- Many classical statistical learning methods operate in the supervised learning domain
 - *Example*: linear regression, logistic regression, ...

In **unsupervised learning**:

- We have observations x_i but no response y_i
- “Unsupervised” because there is no response to guide the analysis
- Often used to explore relationships among observations or variables
 - *Example*: Clustering, dimension reduction, ...

Some problems lie between these two categories and the distinction is not always sharp

Illustration: Supervised vs. unsupervised learning

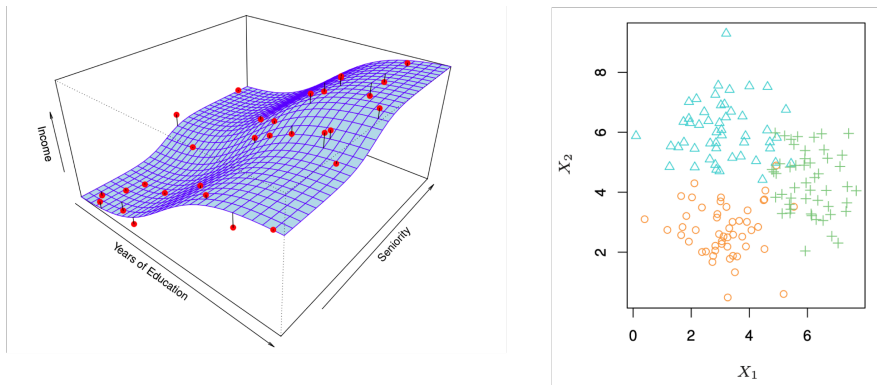


Figure: Supervised vs. unsupervised learning

For now, we focus on supervised learning

Regression vs. classification

Variables can be quantitative or qualitative (categorical):

- Quantitative variables take numeric values
- Qualitative variables belong to one of K different *classes*

Depending on whether the **response** is quantitative or qualitative:

- **Regression:** Supervised learning problems with a *quantitative* response
- **Classification:** Supervised learning problems with a *qualitative* response

The response type (not the method) determines the task:

- e.g., logistic regression is a classification method because the response is qualitative

Predictors may be quantitative or categorical in either task

Illustration: Regression vs. classification

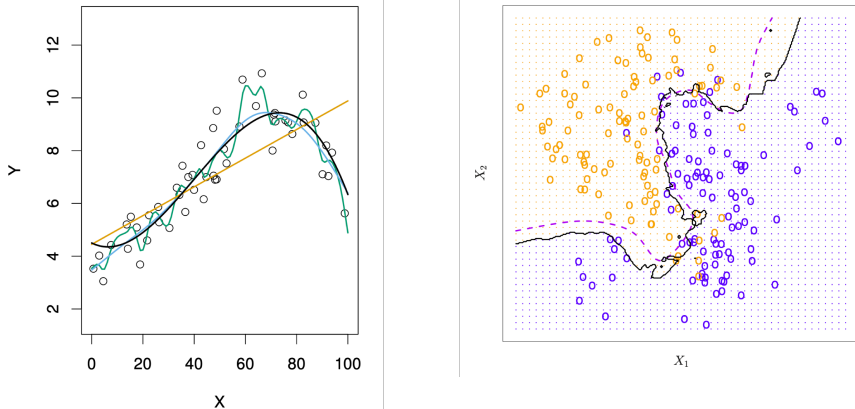


Figure: Regression vs. classification

Regression: Key questions

Regression problems are supervised learning problems with a *quantitative* response

Typical questions we want to address via regression include:

- Is there any relationship between X and Y ?
- How strong is it? (How much of Y is explained by X ?)
- How large is the association? (How does Y change per unit increase in X ?)
- How accurately can we predict Y given X ?
- Is the relationship linear?

With multiple predictors, we can also ask (in future lectures):

- Which X are associated with Y ?
- Are there interactions among X ?

Example: Advertising data

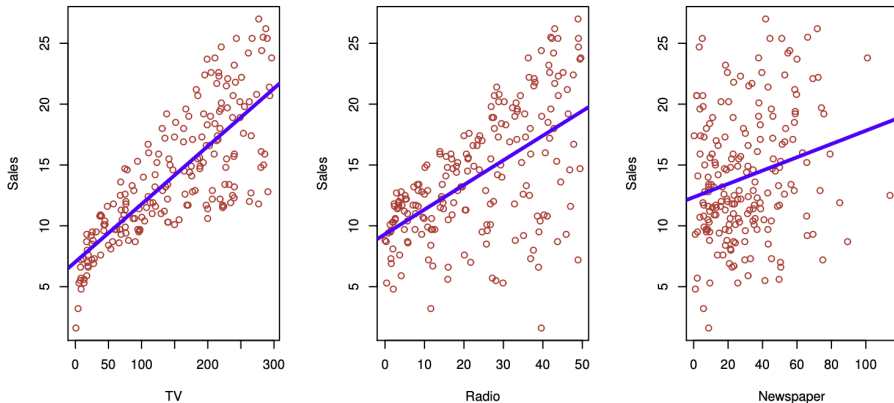


Figure: The Advertising data set shows Sales of a product in 200 different markets against advertising budgets for three media: TV, Radio, and Newspaper [JWHT21, Figure 2.1].

For simple linear regression, we focus on one predictor at a time

Simple linear regression

Simple linear regression models the mean response using a single predictor X , assuming an approximately linear relationship between X and Y

Mathematically, we assume

$$Y = \beta_0 + \beta_1 X + \epsilon,$$

- *Model parameters:* β_0 (intercept), β_1 (slope) are fixed, unknown constants
- ϵ is an error term that captures noise and other unmodeled factors, with $\mathbb{E}[\epsilon \mid X] = 0$

We often say “we *regress* Y on X ”

Once we have *estimated* $\hat{\beta}_0$ and $\hat{\beta}_1$ from training data, we can predict

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

Estimating the coefficients: Least squares

In practice, β_0 and β_1 are unknown and must be estimated from data

$$(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$$

We choose $\hat{\beta}_0$ and $\hat{\beta}_1$ so that the fitted values are close to the observed y_i :

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i \approx y_i$$

The most common approach involves the **least squares** criterion:

- For candidate coefficients b_0, b_1 , the residual sum of squares is

$$\text{RSS}(b_0, b_1) = \sum_{i=1}^n (\hat{y}_i - y_i)^2 = \sum_{i=1}^n (b_0 + b_1 x_i - y_i)^2$$

- The *least squares* approach chooses $\hat{\beta}_0$ and $\hat{\beta}_1$ that minimize the RSS:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} \quad \text{and} \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

where $\bar{y} := \frac{1}{n} \sum_{i=1}^n y_i$ and $\bar{x} := \frac{1}{n} \sum_{i=1}^n x_i$ (Exercise: derive these, cf. Homework 3-(a))

Pop-up quiz: Least squares coefficients

Dataset: Three observations:

$$(x_1, y_1) = (1, 2), \quad (x_2, y_2) = (2, 3), \quad (x_3, y_3) = (3, 6).$$

Question: What are the estimated slope $\hat{\beta}_1$ and intercept $\hat{\beta}_0$?

Hints (partial sums to speed up):

$$\bar{x} = 2, \quad \bar{y} = \frac{2 + 3 + 6}{3} = \frac{11}{3}, \quad \sum (x_i - \bar{x})^2 = 2, \quad \sum (x_i - \bar{x})(y_i - \bar{y}) = 4.$$

- a) $\hat{\beta}_1 = 2, \quad \hat{\beta}_0 = -\frac{1}{3}$
- b) $\hat{\beta}_1 = 1.5, \quad \hat{\beta}_0 = 1$
- c) $\hat{\beta}_1 = 2, \quad \hat{\beta}_0 = 0$
- d) $\hat{\beta}_1 = 3, \quad \hat{\beta}_0 = -2$

Answer: The correct choice is **A** because

$$\hat{\beta}_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{4}{2} = 2, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = \frac{11}{3} - 2 \cdot 2 = -\frac{1}{3}$$

Properties of the least squares estimator

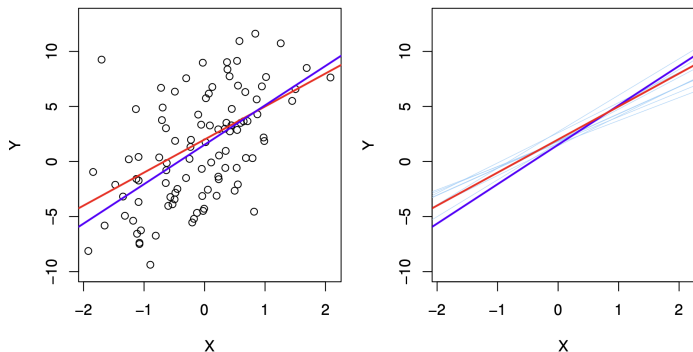


Figure: Least squares coefficient estimates from 100 data points. Red: population regression line, Blue: least squares line, Light blue: ten separate least squares lines [JWHT21, Figure 3.3].

$(\hat{\beta}_0, \hat{\beta}_1)$ estimate (β_0, β_1) using (random) data, so different data produce different fits
 $\implies \hat{\beta}_0$ and $\hat{\beta}_1$ are random variables

Properties of the least squares estimator (cont'd)

Unbiasedness: For fixed predictor values x_i , under the linear model with $\mathbb{E}[\epsilon_i] = 0$, $\hat{\beta}_0$ and $\hat{\beta}_1$ are *unbiased* estimators of β_0 and β_1

- $\mathbb{E}[\hat{\beta}_0] = \beta_0$ and $\mathbb{E}[\hat{\beta}_1] = \beta_1$
- Across independent repeated samples at these x_i , averages of the estimates approach the true coefficients

Sampling distribution: Nevertheless, we only have one dataset!

- We care about how far $\hat{\beta}_0, \hat{\beta}_1$ can deviate from their expected values
- Under certain assumptions¹, the standard errors of $\hat{\beta}_0$ and $\hat{\beta}_1$ are:

$$\text{SE}(\hat{\beta}_0)^2 = \sigma^2 \left[\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right] \quad \text{and} \quad \text{SE}(\hat{\beta}_1)^2 = \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

where $\sigma^2 = \text{Var}(\epsilon)$, which is typically **unknown and has to be estimated**

¹These formulas require that the errors ϵ_i have mean zero, common variance σ^2 , and are uncorrelated

Inference about the model parameters

For $n > 2$, we estimate $\sigma = \sqrt{\text{Var}(\epsilon)}$ using the *residual standard error* (RSE):

$$\hat{\sigma} = \text{RSE} = \sqrt{\frac{\text{RSS}}{n-2}}$$

where $\text{RSS} = \sum_{i=1}^n (\hat{y}_i - y_i)^2 = \sum_{i=1}^n (\hat{\beta}_0 + \hat{\beta}_1 x_i - y_i)^2$

Confidence Intervals: Assume independent $\epsilon_i \sim N(0, \sigma^2)$, $\sigma^2 > 0$

- A 95% confidence interval for β_j , $j = 0, 1$, is² $\hat{\beta}_j \pm t_{n-2, 0.975} \widehat{\text{SE}}(\hat{\beta}_j)$, approximately

$$\left[\hat{\beta}_j - 1.96 \cdot \widehat{\text{SE}}(\hat{\beta}_j), \hat{\beta}_j + 1.96 \cdot \widehat{\text{SE}}(\hat{\beta}_j) \right] \quad (\text{when } n \gg 1)$$

- In repeated sampling, about 95% of these intervals contain the true β_j

² $t_{\nu, 0.975}$ is the 97.5th percentile of t_{ν} ; $\widehat{\text{SE}}$ substitutes $\hat{\sigma}$ for σ in the SE formula.

Inference about the model parameters (cont'd)

Hypothesis Testing: Standard errors can also be used for hypothesis tests on β_0 or β_1

- A common test:

$$H_0 : \beta_1 = 0 \quad (\text{zero slope}) \quad \text{vs.} \quad H_1 : \beta_1 \neq 0 \quad (\text{nonzero slope})$$

- Under H_0 and the preceding Gaussian-error assumptions, the test statistic follows a t -distribution with $n - 2$ degrees of freedom:

$$t = \frac{\hat{\beta}_1 - 0}{\widehat{\text{SE}}(\hat{\beta}_1)}$$

- * Large $|t|$ gives stronger evidence against H_0 (equivalently, a smaller p-value).

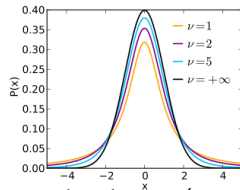


Figure: t -distribution (image)

Pop-up quiz: Standard error and hypothesis test

Let's use the *same dataset* as before:

$$(1, 2), (2, 3), (3, 6),$$

We found $\hat{\beta}_1 = 2$ and $\hat{\beta}_0 = -\frac{1}{3}$; also observe that $\sum (x_i - \bar{x})^2 = 2$

1) **Residual standard error:** We have $n = 3$, so $n - 2 = 1$. The RSS is

$$\text{RSS} = \sum_{i=1}^3 (y_i - \hat{y}_i)^2 = \frac{2}{3} \quad \Rightarrow \quad \hat{\sigma} = \sqrt{\frac{\text{RSS}}{n-2}} = ?$$

2) **Standard error of $\hat{\beta}_1$:** $\widehat{\text{SE}}(\hat{\beta}_1) = \sqrt{\frac{\hat{\sigma}^2}{\sum (x_i - \bar{x})^2}} = ?$

3) **Hypothesis test:** $H_0 : \beta_1 = 0$ vs. $H_1 : \beta_1 \neq 0$. What is the t -statistic $t = \frac{\hat{\beta}_1 - 0}{\widehat{\text{SE}}(\hat{\beta}_1)} = ?$

A) $\hat{\sigma} \approx 0.82$, $\widehat{\text{SE}}(\hat{\beta}_1) \approx 0.58$, $t \approx 3.46$

B) $\hat{\sigma} \approx 1.0$, $\widehat{\text{SE}}(\hat{\beta}_1) \approx 0.5$, $t \approx 2.0$

C) $\hat{\sigma} \approx 0.58$, $\widehat{\text{SE}}(\hat{\beta}_1) \approx 0.82$, $t \approx 0.71$

D) $\hat{\sigma} \approx 0.50$, $\widehat{\text{SE}}(\hat{\beta}_1) \approx 0.25$, $t \approx 8.0$

Pop-up quiz: Standard error and hypothesis test

Let's use the *same dataset* as before:

$$(1, 2), (2, 3), (3, 6),$$

We found $\hat{\beta}_1 = 2$ and $\hat{\beta}_0 = -\frac{1}{3}$; also observe that $\sum (x_i - \bar{x})^2 = 2$

Answer: Since $RSS = \frac{2}{3}$ and $n - 2 = 1$,

$$\hat{\sigma} = \sqrt{\frac{RSS}{n-2}} = \sqrt{\frac{2}{3}} \approx 0.816$$

Also,

$$\widehat{SE}(\hat{\beta}_1) = \sqrt{\frac{\hat{\sigma}^2}{\sum (x_i - \bar{x})^2}} = \sqrt{\frac{2/3}{2}} = \sqrt{\frac{1}{3}} \approx 0.577$$

Therefore,

$$t = \frac{\hat{\beta}_1}{\widehat{SE}(\hat{\beta}_1)} = \frac{2}{\sqrt{1/3}} = 2\sqrt{3} \approx 3.46$$

So the correct choice is **A**

* Note: with one degree of freedom, $t_{1,0.975} \approx 12.71$: $3.46 < 12.71$, so we do **not** reject H_0 at the 5% level

Assessing the accuracy of the model

The quality of an in-sample linear fit is summarized using RSE and the R^2 statistic

- **Residual standard error (RSE):** “*typical deviation of Y from the regression line*”

$$\text{RSE} = \sqrt{\frac{\text{RSS}}{n-2}} = \sqrt{\frac{1}{n-2} \sum_{i=1}^n (y_i - \hat{y}_i)^2},$$

which is an estimate of the standard deviation of ϵ

- **The R^2 :** “the proportion of observed variation captured by the fitted model”

$$R^2 = \frac{\text{TSS} - \text{RSS}}{\text{TSS}} = 1 - \frac{\text{RSS}}{\text{TSS}}$$

where $\text{TSS} = \sum_{i=1}^n (y_i - \bar{y})^2$ is the total sum of squares (TSS); assume $\text{TSS} > 0$

The R^2 statistic

For least squares with an intercept and $\text{TSS} > 0$, training R^2 is scale-free and lies in $[0, 1]$

- R^2 near 1 indicates most variability in Y is explained by the linear regression model
- R^2 near 0 indicates little variability is explained
 - e.g., weak linear signal relative to noise, or a nonlinear relationship

Recall the (sample) correlation:

$$\widehat{\text{Corr}}(X, Y) = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}$$

- In simple linear regression, $R^2 = r^2$, where $r = \widehat{\text{Corr}}(X, Y)$ is the sample correlation
- With multiple predictors, R^2 is generally not the squared correlation of Y with any single predictor
- A high training R^2 does not guarantee good test performance

Wrap-up

Summary:

- Regression is a supervised learning problem with a quantitative response
- Simple linear regression models the mean response as

$$\mathbb{E}[Y \mid X] = \beta_0 + \beta_1 X$$

- Least squares estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ by minimizing the residual sum of squares (RSS)
- Standard errors, confidence intervals, and t -tests quantify uncertainty in the fitted coefficients
- Model fit is commonly summarized by the residual standard error (RSE) and R^2

Suggested reading: [JWHT21, Ch. 3.1]

Next lecture: Multiple linear regression

References



Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani.

An Introduction to Statistical Learning: with Applications in R, volume 112 of *Springer Texts in Statistics*.

Springer, New York, NY, 2nd edition, 2021.